

Tropical Cyclone Genesis Theories-II

Conceptual framework and underlying instability

Lecture notes from Muller et al. (2022), Muller and Romps (2018),
and Tang et al. (2020)

MEA 458/NCSU

Learning goals

By the end of the lecture, students should be able to:

- ▶ Be able to describe the conceptual idea behind convective-self aggregation
- ▶ Be able to describe, in words, the underlying instability
- ▶ Be able to describe the conceptual idea behind the Marsupial Pouch Theory for TC genesis

Central question

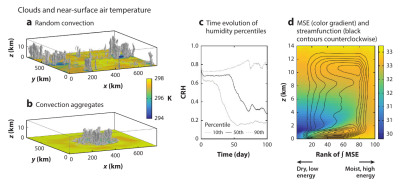
Can tropical cyclogenesis be viewed partly as **rotating convective self-aggregation**?

Not that all TC genesis is the outcome of self-aggregation (clearly we can see pre-existing seed disturbances with organized convection). It is that some feedbacks identified in idealized self-aggregation also help precondition, organize, and accelerate the formation of a tropical cyclone.



What is convective self-aggregation?

- In idealized radiative-convective equilibrium (RCE), convection can spontaneously cluster even with homogeneous SST, forcing, and boundary conditions.
- The final state separates the domain into a moist convective region and broad dry subsiding regions.
- The organizing variable is often column humidity or vertically integrated moist static energy.



Source: Muller et al. (2022), Fig. 1

Moist Static Energy: “Fuel” for Tropical Convection

Definition

Moist static energy (MSE) combines three forms of energy carried by an air parcel:

$$h = c_p T + gz + L_v q$$

- ▶ $c_p T$: **thermal energy**
Warmer air has more energy.
- ▶ gz : **potential energy**
Air at greater height has more gravitational potential energy.
- ▶ $L_v q$: **latent energy**
Water vapor stores energy that can be released when condensation occurs.

Why is MSE useful?

For approximately moist-adiabatic motion,

$$h \approx \text{constant.}$$

So MSE lets us follow the exchange between

heat $\longleftrightarrow$ water vapor $\longleftrightarrow$ height.

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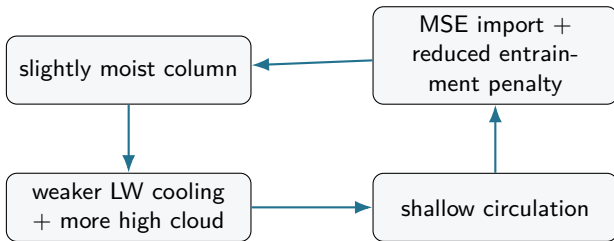
In the tropics

High MSE near the surface generally means air is **warm and moist** and therefore favorable for deep convection.

Think of MSE as the combined thermodynamic energy available to moist tropical air.

Self Aggregation: The underlying instability

A small moist anomaly can amplify when radiation, convection, and circulation move moist static energy **upgradient**: from dry, low-energy columns into moist, high-energy columns.



This is an instability of spatial variance: moist columns become moister and dry columns become drier, unless mixing, surface fluxes, or other processes arrest the growth.

Moist static energy variance budget

The standard diagnostic asks whether a process increases the variance of column MSE:

$$\frac{1}{2} \frac{d}{dt} \left(\widehat{\text{MSE}}' \right)^2 = \left\langle \widehat{\text{MSE}}' F'_{sfc} \right\rangle + \left\langle \widehat{\text{MSE}}' Q'_{rad} \right\rangle - \left\langle \widehat{\text{MSE}}' C'_h \right\rangle.$$

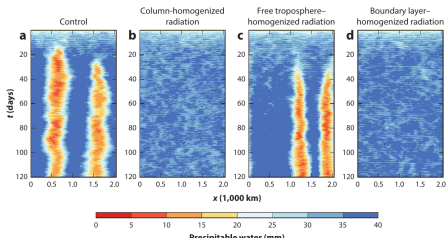
- ▶ Positive covariance between high MSE and anomalous heating or surface fluxes amplifies organization.
- ▶ Radiative cooling in dry columns and reduced cooling/heating in cloudy columns create differential buoyancy forcing.
- ▶ The induced shallow circulation exports near-surface high-MSE air from dry regions into moist regions, producing an upgradient energy transport.

Interpretation

A positive feedback is not simply “more heating.” It is heating, cooling, or advection that is spatially phased to increase MSE variance.

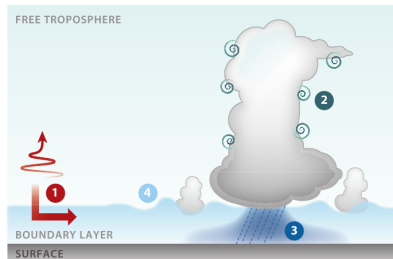
Mechanism-denial evidence: radiation matters

- ▶ The control simulation develops persistent moist and dry regions.
- ▶ Horizontally homogenizing radiation strongly suppresses aggregation.
- ▶ Boundary-layer radiative feedbacks are especially important because buoyancy anomalies can persist there more readily than in the free troposphere.



Source: Muller et al. (2022), Fig. 2

A hierarchy of physical feedbacks



Source: Muller et al. (2022), Fig. 4

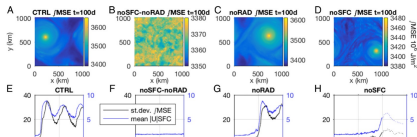
- ① Radiation: enhanced cooling in dry regions drives shallow divergent flow.
- ② Entrainment: moist surroundings reduce dilution of updrafts.
- ③ Cold pools: gust fronts trigger new convection and concentrate vorticity.
- ④ Boundary-layer waves: convection emits gravity waves that organize later convection.

Different models emphasize different members of this hierarchy. The robust point is the positive coupling among humidity, convection, radiation, and low-level convergence.

Rotating RCE: aggregation becomes cyclogenesis

STATIONARY TROPICAL WAVE (TROPICAL) MODELS: THE HIGHEST-ORDER VERTICAL INTEGRAL can be changed only by radiation, surface fluxes, and advection. In the tropics, where horizontal temperature gradients are small, its variability is closely related to that of water

UNIFORMITY OF THE DEEP L.T.P.C., BUT SURFACE ENERGY AND HIGH PERTURBATIONS ON PRECIPITABLE WATER AND LATENT HEAT FLUX). This is again consistent with our current understanding of TCs whose main sources of energy in the mature stage are interactive surface fluxes (14).



Source: Muller and Roms (2018), Fig. 1

- ▶ With rotation, the moist aggregate can acquire a coherent cyclonic circulation.
- ▶ In the control run, a TC forms in about 25–30 days.
- ▶ Removing both radiative and surface-flux feedbacks prevents genesis in this idealized setup.
- ▶ Removing radiation alone delays genesis by more than a factor of two.

Feedbacks that make aggregation grow

Self-aggregation is maintained by feedbacks that sharpen moist–dry contrasts.

Radiation–moisture feedback

Dry regions cool radiatively and subside; moist regions cool less and remain convectively active. This reinforces the dry/moist contrast.

Surface-flux feedback

Convective circulations can enhance near-surface winds and surface enthalpy fluxes into the moist region.

Entrainment feedback

Convection is more efficient in humid air because entrainment dilutes updrafts less strongly.

Cold-pool and wave feedbacks

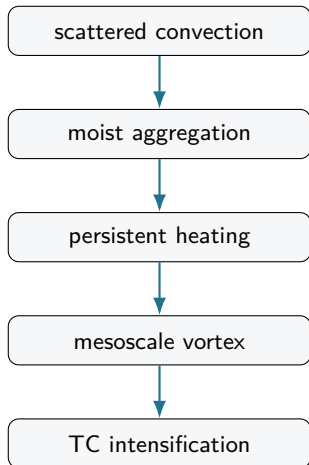
Cold pools and gravity waves can help trigger nearby convection, but they can also spread convection and delay aggregation, depending on the setup.

Wing and Emanuel (2014); Jeevanjee and Romps (2013); Wing et al. (2017); Muller et al. (2022).

Why self-aggregation can accelerate cyclogenesis

When rotation is present, an aggregated convective cluster can become a favorable precursor to tropical cyclogenesis.

- ▶ The moist cluster reduces dry-air entrainment into deep convection.
- ▶ Persistent heating produces a mesoscale pressure and circulation response.
- ▶ Convective-scale vorticity anomalies are repeatedly generated near one another and can merge upscale.



Muller et al. (2018) found that self-aggregation can substantially shorten the time to genesis in idealized rotating simulations.

Self-aggregation: conceptual caveats

The self-aggregation framework is powerful, but its direct application to real tropical cyclogenesis requires care.

Idealized origin

Most self-aggregation studies begin from homogeneous SST and weak or absent imposed large-scale organization.

Mechanism dependence

The initiating mechanism need not be the same as the maintenance mechanism. Radiation, cold pools, surface fluxes, and rotation can change the pathway.

Time scale

Aggregation in many simulations occurs over days to weeks, whereas real genesis often occurs inside pre-existing waves, monsoon gyres, or convective envelopes.

Interpretation

In nature, self-aggregation is best viewed as a set of feedbacks that can reinforce organization, not necessarily as a complete standalone genesis theory.

TC Formation Views: Marsupial Paradigm

The Marsupial Paradigm View

Link: Dunkerton, Montgomery, and Wang (2009).

Marsupial pouch: the basic idea

Conceptual shift

Tropical cyclogenesis is not viewed simply as convection occurring inside a broad easterly wave. It is viewed as development inside a **co-moving, semi-closed Lagrangian pouch** embedded in the wave.

The favored location is the **“sweet spot”**: the intersection of the wave trough and the critical latitude, where

$$\bar{u}(y_c) \approx c,$$

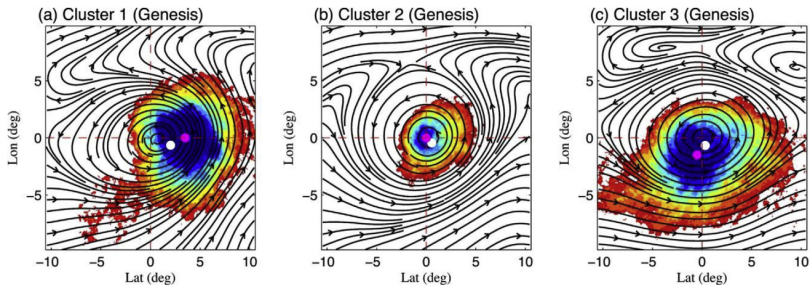
with c the phase speed of the parent wave.

Dunkerton, Montgomery, and Wang (2009).

Marsupial pouch: the basic idea

Conceptual shift

Tropical cyclogenesis is not viewed simply as convection occurring inside a broad easterly wave. It is viewed as development inside a **co-moving, semi-closed Lagrangian pouch** embedded in the wave.



Source: Tang et al. (2020), Fig. 6; from Wang (2018)

Why the pouch is a favorable incubator

The pouch paradigm emphasizes three mutually reinforcing properties.

1. Protection

Closed or nearly closed streamlines reduce ventilation by dry environmental air and lessen the disruptive effects of shear-relative inflow.

2. Repeated moistening

Air parcels remain near the convective region, so convection progressively humidifies the lower and middle troposphere.

3. Vorticity gathering

Wave breaking and roll-up concentrate low-level cyclonic vorticity near the critical layer.

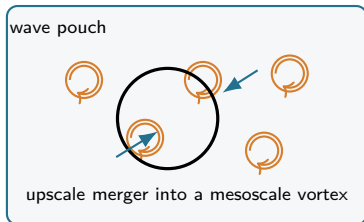


This is the “marsupial” analogy: the wave pouch carries and protects the developing disturbance.

How the proto-vortex forms inside the pouch

Inside the pouch, deep convection does more than release latent heat. It also produces and organizes rotating convective elements.

- ▶ Convective bursts stretch ambient cyclonic vorticity.
- ▶ Mesoscale vortices merge upscale through vortex interaction.
- ▶ Diabatic heating amplifies the parent wave and helps build a positive potential-vorticity anomaly.
- ▶ As the environment moistens, convection becomes less intermittent and more spatially focused.



From pouch-scale disturbance to tropical cyclone

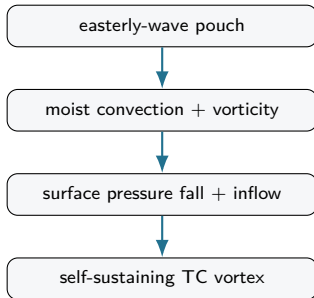
The pouch supplies an organized, moist, cyclonic starting point; it does not by itself guarantee genesis.

Necessary transition

The seed must become vertically coherent enough for surface pressure falls, boundary-layer inflow, and surface enthalpy flux feedbacks to reinforce the circulation.

In this view, genesis is a **protected bottom-up organization**

problem: convection and low-level vorticity remain collocated until the vortex couples to the boundary layer.



Side-by-side synthesis

	Marsupial pouch	Convective self-aggregation
Starting point	Pre-existing tropical wave and critical layer	Homogeneous or weakly forced convection in RCE
Organizer	Wave-relative closed circulation	Internal convection–moisture–radiation–flux feedbacks
Protected region	Pouch near wave trough/critical latitude	Moist convective cluster surrounded by dry subsidence
Main TC relevance	Keeps the seed moist, coherent, and shielded while vorticity aggregates	Builds a moist, persistent convective region that can spin up under rotation
Best mental model	<i>A protected incubator carried by a wave</i>	<i>A clustering instability of moist convection</i>

Common thread: tropical cyclogenesis becomes more likely when convection, moisture, and cyclonic vorticity remain collocated long enough to build a coherent lower-tropospheric vortex.

Summary

- ① Self-aggregation is a spontaneous instability of convection, moisture, radiation, and circulation in RCE.
- ② The key positive feedback is upgradient MSE transport into moist convective regions, supported strongly by LW radiative contrasts.
- ③ In rotating RCE, the aggregated moist region can become a tropical cyclone; radiation accelerates genesis by organizing the precursor state.
- ④ In observed genesis, aggregation feedbacks operate within larger-scale constraints: pouch protection, moisture preconditioning, frictional convergence, vertical alignment, and environmental shear.

References and figure sources

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